



Example Hotel

Example Site / Example Location

Net Zero Discovery Assessment Report

Second Discovery Assessment – Progress Review & New Recommendations

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Date: 15/01/2025 **Previous Assessment:** DD/MM/YYYY

Executive Summary

This second Discovery Assessment reviewed progress at Example Hotel since the first visit in January 2024, compared energy consumption between the two periods, and identified a further set of energy efficiency and renewable energy opportunities. The key findings are summarised below; full detail follows in the rest of this report.

 12.3% Gas consumption (vs Year 1 baseline)	 9.1% Electricity consumption (vs Year 1 baseline)	 10.8% Carbon emissions (CO ₂) (vs Year 1 baseline)
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Watch Point: Overnight Base Load

Electricity consumption between midnight and 6am remains disproportionately high relative to room occupancy. Significant savings are available through improved HVAC scheduling, kitchen extract controls and laundry equipment timers.

What's Changed Since January 2024

- LED lighting upgrade completed across all guest bedrooms and corridors – estimated 18% electricity saving on lighting circuits.
- Pool heat pump serviced and refrigerant topped up – improving COP from 2.8 to 3.6.
- Voltage optimisation unit installed on main kitchen circuit.
- Staff energy awareness programme launched, with energy champions appointed on each floor.

Top 5 Priority Recommendations

1. EnPact Filter Pro – power quality improvement at the mains incomer (10–15% electricity saving)
2. Insulation Jackets – boiler plant room and hot water storage (circa 10% gas saving)
3. Limescale Inhibitors – AquaVantage across all 87 guest room shower circuits
4. Real-Time Energy Monitoring – HVAC, mains incomer, kitchen and laundry
5. Gas Saving Solutions on Boilers – EndoTherm and magnetic fuel conditioners (10–15% gas saving)

See 'Top 5 Priority Recommendations' for full explanations, indicative costs and expected ROI for each.

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Introduction

On Wednesday 15th January 2025, Jon Cullum returned to Example Hotel for a second Net Zero Discovery Assessment – twelve months after the first visit in January 2024. This report reviews progress against the 2024 recommendations, benchmarks energy use year-on-year, and sets out the next phase of energy efficiency and carbon reduction opportunities.

Attendees

- Green Wing Energy Solutions – Jon Cullum
- Example Hotel – Example Contact Name, Head of Facilities & Engineering

Areas of Focus

- Review of progress against the January 2024 recommendations
- Energy benchmarking and year-on-year comparison (gas, electricity and carbon)
- Boiler plant room controls and hot water pipework insulation
- Mains incomer power quality and circuit-level energy monitoring
- HVAC controls and air handling units – scheduling and occupancy settings
- Guest room and corridor lighting – post-LED upgrade review
- Kitchen and laundry base load – overnight consumption profile
- Swimming pool plant – heat pump, filtration and water treatment

Property Overview

Client	Example Hotel
Site Name	Example Hotel, Example Location
Site Contact	Example Contact Name – Head of Facilities & Engineering
Building Use	4-star hotel – 120 guest rooms, restaurant, bar, conference suite, spa and swimming pool
Operating Hours	24/7 – 365 days per year
Boilers	3 × Ideal EvoMax 120 condensing gas boilers
Air Conditioning	Central HVAC with 6 air handling units; fan coil units in guest rooms
Lighting	LED completed across rooms and corridors; remaining legacy fittings in plant areas
BMS	Trend BMS – present and actively managed
EPC Rating	Current: D – Target: C (short term), B (medium term)
Energy Scope	Gas, Electricity, Water
Sustainable Energy	None currently – solar PV feasibility under review

Energy Benchmarking – Year-on-Year Comparison

To track progress, we have compared energy consumption for the twelve months following the first Discovery Assessment (February 2024 – January 2025) against the benchmark period from the first report (January 2023 – December 2023).

Metric	Year 1 (Jan 2023 – Dec 2023)	Year 2 (Feb 2024 – Jan 2025)	Change
Gas	1,842,330 kWh	1,615,560 kWh	▼ 12.3%
Electricity	687,120 kWh	624,490 kWh	▼ 9.1%
Carbon (CO ₂)	452.1 tonnes	403.3 tonnes	▼ 10.8%

Both gas and electricity consumption have reduced year-on-year, and total carbon emissions are down by over 10%. This reflects the impact of the LED lighting upgrade, the pool heat pump service and improved staff engagement on energy behaviours since the first assessment.

Watch Point: Overnight Base Load

While annual consumption is trending down, analysis of half-hourly data shows electricity base load between midnight and 6am has not reduced proportionally. The overnight period represents approximately 22% of total annual electricity consumption – disproportionately high for a property at average 60% occupancy. Improved HVAC scheduling, kitchen extract controls and laundry timers are the primary opportunities.

Top 5 Priority Recommendations

Based on discussions and observations throughout this Discovery Assessment, the following five measures represent the highest-impact opportunities currently available to Example Hotel.

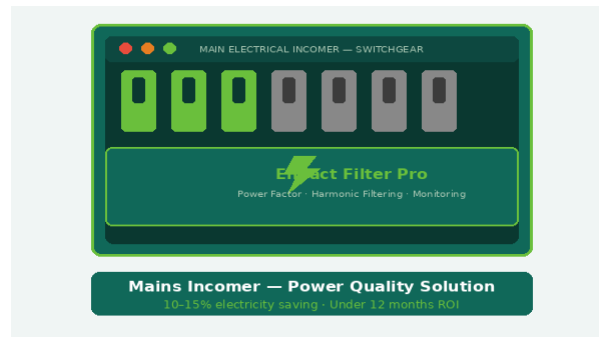
#	Recommendation	Indicative Cost	Estimated Saving	Expected ROI	Saving Type
1	EnPact Filter Pro	circa £10,000	10–15% electricity	Under 12 months	Electrical
2	Insulation Jackets – Boiler Plant	Free survey; cost TBC	circa 10% gas	Circa 12 months	Gas
3	Limescale Inhibitors – AquaVantage	circa £18,500	Reduced water heating & maintenance	Under 12 months	Electrical & Gas
4	Real-Time Energy Monitoring	Included with EnPact	Enables & verifies other savings	Circa 12 months	Electrical
5	Gas Saving Solutions on Boilers	At next boiler service	10–15% gas	Circa 12 months	Gas

1. EnPact Filter Pro

EnPact Filter Pro is a passive power conditioning unit installed at the building's main electrical incomer. It corrects power factor, filters out electrical harmonics and stabilises voltage delivered to circuits throughout the building – helping motors, pumps, HVAC units and laundry equipment run more efficiently, cooler and for longer. Given the mix of HVAC plant, pool pumps and commercial kitchen equipment at the hotel, this is one of the most cost-effective electrical upgrades available.

- Already proposed to Example Hotel – proposal available on request.
- Recommended Solution Partner: Emissis

Main electrical intake room – where EnPact Filter Pro would be fitted.

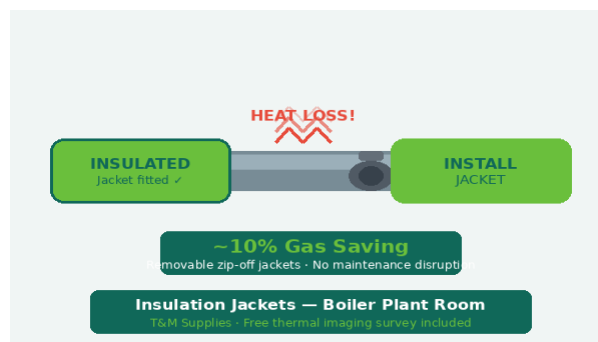


2. Insulation Jackets – Boiler Plant Room

Customised, removable insulation jackets fit directly over pipework, valves, flanges and heat exchangers in the boiler plant room. Unlike fixed lagging, they unzip for maintenance access and refit afterwards. Every exposed section of hot pipework currently loses heat into the plant room rather than delivering it to guest rooms, the pool or catering – making this one of the simplest ways to reduce gas consumption with a rapid payback.

Boiler plant room – exposed pipework and uninsulated valves throughout.

- All three boilers showed exposed pipework and uninsulated valves during this visit – a free survey is the recommended next step.
- Recommended Solution Partner: T&M Supplies

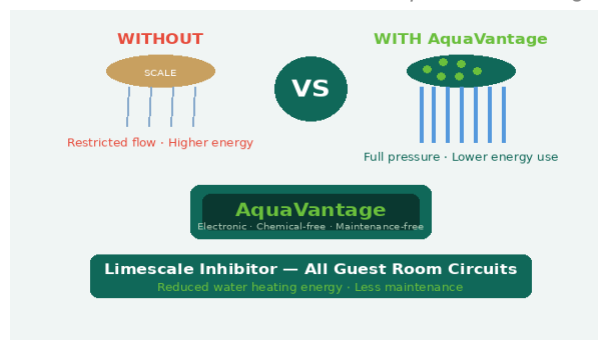


3. Limescale Inhibitors – AquaVantage

AquaVantage is an electronic, chemical-free water conditioning system fitted to the incoming water supply. It prevents new limescale forming on shower heads, pipework and water heating surfaces across all 87 guest rooms – scale that currently restricts flow, insulates heating elements and pushes up the energy needed to heat water. Removing this barrier cuts both electrical and gas consumption for water heating while reducing ongoing maintenance disruption.

- 87 guest room shower circuits affected – significant ongoing maintenance cost.
- Recommended Solution Partner: AquaVantage

Guest room shower – limescale build-up visible on fittings.

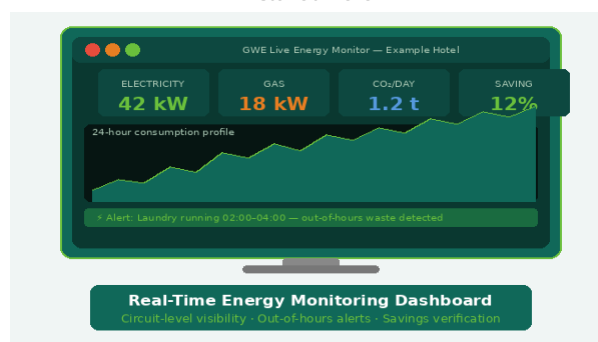


4. Real-Time Energy Monitoring

Circuit-level monitoring sensors fitted to key circuits and incomers – mains electricity, gas, water, HVAC, kitchen and laundry – feed live data to an online dashboard. This gives the facilities team visibility of exactly where, when and how energy is being used, including the overnight base load profile identified in this assessment. It provides the before-and-after data needed to confirm savings from all other measures.

- Covers HVAC, mains incomer, kitchen, laundry and all plant rooms.
- Recommended Solution Partner: Emissis

Main distribution board – circuit-level monitoring to be installed here.

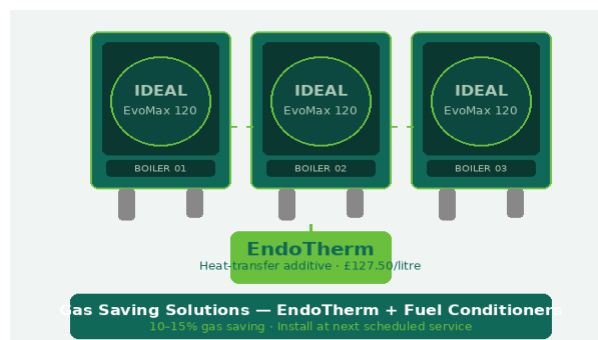


5. Gas Saving Solutions on Boilers

EndoTherm is a heat-transfer additive dosed into the boiler heating circuit, improving how efficiently heat moves from water into radiators and heat exchangers – allowing the boilers to deliver the same output at a lower flow temperature or shorter run time, typically reducing gas consumption by up to 15%. Magnetic fuel conditioners, fitted to the gas supply lines, improve combustion efficiency at the burners. Both are best installed at the next scheduled boiler service.

- Three Ideal EvoMax 120 boilers – EndoTherm dose at next service; price: £127.50/litre.
- Recommended Solution: EndoTherm (endothem.co.uk)

Boiler plant room – three Ideal EvoMax 120 condensing boilers.



Other Solutions Referenced in This Report

EnPact HVAC & Contine WM

EnPact HVAC is a power conditioning solution designed specifically for air conditioning and HVAC circuits, reducing electrical losses associated with compressor start-up and cycling. Contine WM filters fitted over external coils improve airflow and help units reject heat more efficiently – reducing compressor run-time while filtering dust and allergens. Together, these are recommended for the six AHUs serving the hotel.

- Recommended Solution Installer: Continewm

Building Management System (BMS) Optimisation

The Trend BMS is present and active but operating on schedules set at installation and not reviewed since. An M3 Automation survey is recommended to re-configure HVAC, heating and hot water schedules to match current occupancy patterns – particularly to address the overnight base load identified in the energy data.

- Recommended Solution Installer: M3 Automation Ltd

Technik 2 – Bar & Restaurant Refrigeration

Technik 2 is an energy-saving controller fitted to commercial fridges and bottle coolers. It intelligently manages compressor run-time based on actual cooling demand, reducing electricity consumption without affecting product temperature or food safety. This is recommended for the bottle fridges, cellar cooling and display refrigeration in the bar and restaurant.

- Recommended Solution Partner: Emissis

Detailed Observations & Recommendations

The table below summarises the specific observations made during the site walkaround on 15th January 2025, together with our recommendations, priority rating and the type of saving expected.

Area / Time	Observation	Recommendation	Priority	Saving Type
Boiler Plant Room (09:15)	Three boilers running with basic time controls. Hot water pipework and valves largely uninsulated. No retrofit energy-saving measures in place.	Insulation jackets (T&M Supplies); install EndoTherm and magnetic fuel conditioners at next service.	High	Gas
Mains Incomer (09:35)	No power quality solution at the mains incomer. No circuit-level monitoring in place.	Install EnPact Filter Pro – proposal already provided. Monitoring included.	High	Electrical
AHU Plant Room (09:50)	Six AHUs operating on 2019 schedules. No occupancy-linked controls. Significant out-of-hours runtime observed.	M3 Automation BMS optimisation – reschedule AHUs to match current occupancy patterns.	High	Electrical
Swimming Pool (10:10)	Pool heat pump recently serviced – COP improved. UV treatment system operational. Chemical dosing now automated.	Monitor pool plant energy consumption to confirm savings. Review filtration pump run-times.	Medium	Electrical
Guest Rooms (10:30)	LED upgrade complete. Smart thermostats not yet installed – guest room heating controlled via centralised BMS.	Review BMS guest room setback temperature. Consider smart thermostat pilot on one floor.	Medium	Gas
Kitchen (10:45)	Four commercial fridges and two bottle coolers running continuously. Ovens pre-heating earlier than required. No sub-metering.	Install Technik 2 on fridges/coolers. Add sub-metering. Review oven warm-up schedules with chef.	High	Electrical & Gas
Laundry (11:00)	Three industrial washers running overnight on standard cycles. No economy tariff scheduling in place.	Shift laundry cycles to economy tariff hours where possible. Install monitoring to confirm savings.	High	Electrical
Shower Circuits (11:15)	Limescale build-up visible across sampled guest room shower heads. No water conditioning system in place.	Install AquaVantage across all 87 guest room circuits (circa £18,500, ROI under 12 months).	High	Electrical & Gas

Photo Evidence from This Discovery Assessment

The following photos were taken during the site walkaround on 15th January 2025 and illustrate the key issues identified in the recommendations above.

Photos taken on site will be shown here

Additional Observations from the Site Walkaround

Area	Observation	Recommendation / Note
Conference Suite	Lighting and HVAC left running during unoccupied periods between events.	Link conference suite HVAC and lighting to the BMS occupancy schedule. Consider PIR sensors for lighting.
Spa & Leisure	Pool blanket in place but not consistently used during overnight closure.	Reinforce pool blanket protocol with leisure team to retain heat overnight and reduce heat pump run-time.
EV Charging	No EV charging points currently installed. Car park area suitable.	Commission EV charging feasibility study – strong guest demand and potential revenue stream.
Solar PV	South-facing roof area of approximately 400m ² currently unused.	Commission solar PV feasibility study. Estimated potential: 80–100 kWp, generating circa 80,000 kWh per year.
Staff Awareness	Energy champions appointed but no visible energy dashboards in staff areas.	Install real-time energy display at staff entrance and in the engineering office.
Water Metering	Single site water meter; no sub-metering for laundry, kitchen or spa.	Add sub-metering once energy monitoring is installed to identify further water efficiency opportunities.

Other Recommendations (Carried Forward)

The following recommendations from the first Discovery Assessment remain valid and are carried forward for consideration:

- Solar PV feasibility study – south-facing roof section is well-suited to a 80–100 kWp array
- EV charging installation – car park layout suitable; strong demand from hotel guests
- Rainwater harvesting feasibility for laundry and toilet flushing circuits
- Thermal imaging survey of building fabric, flat roof sections and older window seals
- Continued staff engagement programme with monthly energy KPI reporting at team briefings
- Carbon reduction display at main reception – a strong guest-facing sustainability message

Contact Us

Green Wing Energy Solutions would welcome a discussion on the recommendations of this report and is happy to provide clarification on any questions you may have.

In the first instance, please contact:

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